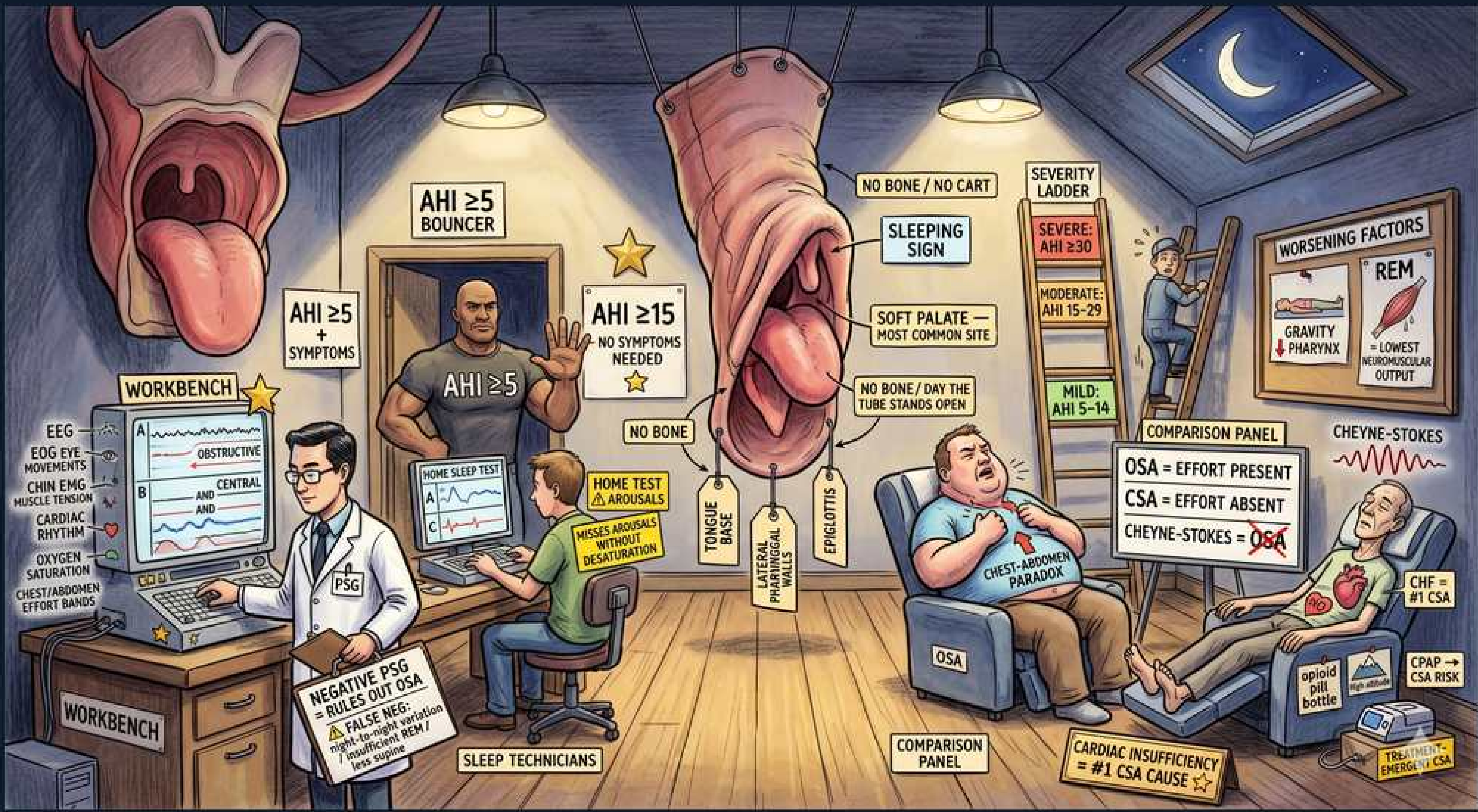
Sleep Apnea — Overview & Pathophysiology



1. Upper-airway collapse

WHAT YOU SEE

pharyngeal airway closing, soft palate collapsing

WHAT IT ENCODES

OSA is the repetitive collapse of the upper (pharyngeal) airway during sleep DESPITE ongoing respiratory effort, causing apneas/hypopneas, cyclic oxygen desaturation, and recurrent cortical arousals that fragment sleep. During sleep the pharyngeal dilator muscles (chiefly the genioglossus) lose tone, so a small or crowded airway collapses on inspiration. The fragmentation produces daytime sleepiness; the intermittent hypoxia plus sympathetic surges drive the cardiovascular complications.

BOARD TRAP

WRONG answer: confusing OSA with central sleep apnea. In OBSTRUCTIVE apnea respiratory EFFORT IS PRESENT (chest/abdomen keep moving, often paradoxically) against a closed airway; in CENTRAL apnea effort is ABSENT (no drive). The discriminator on PSG is the thoracoabdominal effort belts — effort present = obstructive, effort absent = central.

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2. Soft palate — most common site

WHAT YOU SEE

A label pointing at the dangling soft palate / uvula reads SOFT PALATE — MOST COMMON SITE — the floppy, boneless tissue marks the retropalatal pharynx as the #1 collapse point.

WHAT IT ENCODES

The retropalatal pharynx behind the soft palate is the most common site of collapse, followed by the retroglossal region behind the tongue base. The pharynx is the only segment of the airway with NO rigid bony or cartilaginous support, so it depends entirely on dilator-muscle tone, which falls in sleep — explaining why collapse occurs here and not in the rigid trachea or larynx.

BOARD TRAP

WRONG answer: attributing OSA primarily to a fixed nasal or laryngeal obstruction. The dynamic, collapsible RETROPALATAL pharynx is the key site. The discriminator: OSA is a problem of soft-tissue pharyngeal collapse during sleep, which is also why uvulopalatopharyngoplasty (which addresses the soft palate) is only sometimes effective.

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3. Obesity / large neck

WHAT YOU SEE

A heavy-set patient at a WORKBENCH with a thick neck and 'no bone / no cartilage' tags around the airway — the obese body and fat neck literally crowd the collapsible pharynx.

WHAT IT ENCODES

Obesity is the leading risk factor for OSA: parapharyngeal fat narrows the airway and reduced lung volumes lessen tracheal tug/airway tethering. Increased neck circumference is a strong predictor — roughly >43 cm (17 in) in men and >40 cm (16 in) in women. Other risk factors: male sex, increasing age, retrognathia/micrognathia, macroglossia, tonsillar hypertrophy, hypothyroidism, and acromegaly.

BOARD TRAP

WRONG answer: excluding OSA because a patient is not obese. Non-obese patients with craniofacial crowding (retrognathia, large tonsils, macroglossia) can have significant OSA, and OSA also occurs in hypothyroidism/acromegaly. The discriminator: obesity and large neck strongly predict OSA, but a crowded oropharynx (high Mallampati, tonsillar hypertrophy) matters even at normal weight.

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4. Snoring / snore sign

WHAT YOU SEE

A SLEEPING SIGN with a crescent moon hangs over the snoring soft palate — the dangling vibrating tissue under the moon is the snore made visual.

WHAT IT ENCODES

Loud, habitual snoring is the classic nocturnal symptom, produced by turbulent airflow vibrating the relaxed soft palate and pharyngeal walls — often crescendoing then terminating in a silent apnea followed by a gasp/snort. Snoring is highly sensitive (most OSA patients snore) but poorly specific.

BOARD TRAP

WRONG answer: equating snoring with OSA. Simple snoring without apneas/hypopneas, desaturation, or arousals (AHI <5) is NOT OSA. The discriminator: OSA requires documented respiratory events on a sleep study — snoring alone is benign primary snoring until the AHI proves otherwise.

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5. Witnessed apneas

WHAT YOU SEE

The dozing obese man clutching his chest is tagged CHEST-ABDOMEN PARADOX — chest and belly heaving against a shut airway is the witnessed obstructive apnea drawn out.

WHAT IT ENCODES

Bed-partner-witnessed apneas and gasping/choking arousals are among the most SPECIFIC clinical features of OSA. During obstruction the diaphragm and chest wall keep working, producing visible thoracoabdominal PARADOX (chest and abdomen move in opposite directions) as effort builds against the closed airway, until an arousal restores tone and breathing resumes.

BOARD TRAP

WRONG answer: reading thoracoabdominal paradox/continued effort as central sleep apnea. Continued (even increased) effort during the event is the hallmark of OBSTRUCTIVE apnea; central apnea shows a FLAT effort tracing. The discriminator is presence vs absence of respiratory effort during the apneic pause.

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6. Daytime somnolence

WHAT YOU SEE

A heavy man slumped dozing in a chair during the day — the visible daytime nodding-off is the excessive somnolence.

WHAT IT ENCODES

Excessive daytime sleepiness from repeated arousals and sleep fragmentation is the cardinal daytime symptom, along with morning headaches, poor concentration, irritability, and increased motor-vehicle-accident risk. The Epworth Sleepiness Scale (0–24; >10 abnormal) and STOP-BANG questionnaire quantify symptoms and stratify pre-test risk.

BOARD TRAP

WRONG answer: using a high Epworth score (or STOP-BANG) to DIAGNOSE OSA. These are SCREENING tools that raise or lower pretest probability — they cannot confirm or exclude OSA. The discriminator: only a sleep study (PSG or, in selected high-probability patients, HSAT) yields the AHI that establishes the diagnosis.

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7. AHI ≥5 with symptoms

WHAT YOU SEE
A muscular BOUNCER stamped 'AHI ≥5 + SYMPTOMS' guards the door — the bouncer only lets OSA through if events AND symptoms are both present.

WHAT IT ENCODES
OSA is diagnosed at AHI ≥5/hr WITH symptoms/comorbidity (sleepiness, snoring, gasping, hypertension, AF, stroke, diabetes), OR at AHI ≥15/hr regardless of symptoms. An apnea is a ≥90% airflow drop ≥10 s; a hypopnea is a ≥30% airflow drop ≥10 s with desaturation or arousal.

BOARD TRAP
WRONG answer: diagnosing OSA from an AHI of 6 in a completely asymptomatic patient. Between 5 and 14 the symptom/comorbidity requirement is mandatory. The discriminator: AHI 5–14 requires symptoms; AHI ≥15 is diagnostic alone.

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8. AHI ≥15 — no symptoms needed

WHAT YOU SEE
A second bouncer marked 'AHI ≥15 — NO SYMPTOMS NEEDED' wears a GOLD STAR — the gold star flags AHI ≥15 as diagnostic on its own, no symptoms required.

WHAT IT ENCODES
An AHI ≥15 events/hr is diagnostic of OSA even with NO symptoms, because the cardiovascular/metabolic risk at this event burden justifies diagnosis and treatment. This same number 15 is also the floor of the 'moderate' severity band — a deliberately overlapping, frequently tested value.

BOARD TRAP
WRONG answer: deferring diagnosis/treatment in an asymptomatic patient with AHI 20. At ≥15, OSA is established and treatment (typically CPAP) is indicated. The discriminator: ≥15 removes the symptom requirement; below 15 symptoms are mandatory.

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9. Polysomnography (PSG) — gold standard

WHAT YOU SEE
A clinician at a WORKBENCH wired to monitors showing EEG / EOG / EMG / effort-band tracings labelled PSG — the full bench of channels is the gold-standard in-lab sleep study.

WHAT IT ENCODES
In-lab attended polysomnography is the gold-standard diagnostic test, recording EEG/EOG/EMG (sleep staging, arousals), airflow, thoracoabdominal effort, ECG, and oximetry to compute the AHI and classify events as obstructive vs central. It is required when there is significant comorbidity (heart failure, COPD, neuromuscular disease), suspected central apnea/hypoventilation, or when a home study is non-diagnostic.

BOARD TRAP
WRONG answer: trusting a NEGATIVE home sleep test to rule out OSA. Home studies underestimate AHI (no EEG arousals, recording-time denominator) and a negative HSAT in a high-suspicion patient must be followed by full PSG. The discriminator: PSG can both confirm and exclude OSA; HSAT can essentially only confirm it.

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10. Home sleep test caveat

WHAT YOU SEE
A sleep technician at a 'HOME TEST' station with a warning note 'MISSES AROUSALS WITHOUT DESATURATION' — the stripped-down home rig drawn with a false-negative warning.

WHAT IT ENCODES
Home sleep apnea testing is acceptable ONLY for high-probability, uncomplicated moderate-to-severe OSA. It lacks EEG, so it cannot detect arousal-based hypopneas (RERAs) or stage sleep, and it computes AHI over total recording time rather than true sleep time — both biasing AHI downward and risking false negatives.

BOARD TRAP
WRONG answer: using HSAT in a patient with heart failure, suspected central apnea, COPD, hypoventilation, or neuromuscular disease. Those require in-lab PSG. The discriminator is the comorbidity/complexity of the patient — complex cases or a negative HSAT in a high-suspicion patient go to full PSG.

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11. Severity ladder

WHAT YOU SEE
A man climbing a literal SEVERITY LADDER with rungs SEVERE AHI ≥30 / MODERATE 15-29 / MILD 5-14 — each higher rung is a worse AHI band.

WHAT IT ENCODES
Severity by AHI: MILD 5–14, MODERATE 15–29, SEVERE ≥30 events/hr. Severity (with symptoms and comorbidity) guides therapy intensity — moderate-to-severe disease (AHI ≥15, or ≥5 with significant symptoms) is the clearest CPAP indication, while mild disease may respond to weight loss, positional therapy, or an oral appliance.

BOARD TRAP
WRONG answer: misassigning the cut points (e.g., AHI 25 as 'severe'). Memorize 5/15/30. The discriminator: severity bands are mild 5–14, moderate 15–29, severe ≥30 — and CPAP is most strongly indicated at moderate-severe levels.

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12. Worsening factors — REM & supine

WHAT YOU SEE
A WORSENING FACTORS box showing GRAVITY pulling on the pharynx plus a REM tag ('lowest neuromuscular output') — gravity (supine) and REM atonia drawn as the two things that deepen collapse.

WHAT IT ENCODES
REM sleep and the SUPINE position worsen obstructive events. REM produces near-complete skeletal-muscle atonia of the pharyngeal dilators, yielding the longest apneas and deepest desaturations; supine posture lets gravity pull the tongue and soft palate posteriorly. These facts underlie positional therapy and explain REM-predominant OSA.

BOARD TRAP
WRONG answer: calling a study normal when supine/REM AHI is high but overall AHI is borderline (e.g., the patient spent little time supine or in REM). Positional and REM-related OSA is genuine disease. The discriminator: REM atonia and supine gravity cause event clustering, so timing/position of events matters.

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13. Consequences — HTN / pulmonary HTN

WHAT YOU SEE

A panel with a strained heart labelled CHF / PULMONARY HTN — the failing heart is the cardiovascular toll of untreated OSA.

WHAT IT ENCODES

Untreated OSA causes a wide cardiometabolic burden: systemic hypertension (especially RESISTANT and non-dipping), pulmonary hypertension and cor pulmonale (from chronic nocturnal hypoxemia), atrial fibrillation (and post-cardioversion recurrence), coronary disease/MI, stroke, insulin resistance/type 2 diabetes, and increased motor-vehicle accidents. Mechanisms: intermittent hypoxia, sympathetic surges, large negative intrathoracic pressure swings, and inflammation.

BOARD TRAP

WRONG answer: failing to consider OSA in a patient with hypertension resistant to three drugs (including a diuretic). OSA is a leading cause of SECONDARY and resistant hypertension — screen for it. The discriminator: resistant/non-dipping hypertension, new AF, or unexplained pulmonary hypertension in an obese snorer should prompt a sleep study.

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14. OSA vs CSA vs Cheyne-Stokes

WHAT YOU SEE

A COMPARISON PANEL: OSA = EFFORT PRESENT, CSA = EFFORT ABSENT, CHEYNE-STOKES = a crescendo-decrescendo waveform — the three breathing patterns side by side as the effort-present vs effort-absent discriminator.

WHAT IT ENCODES

Obstructive sleep apnea = airflow stops WITH continued respiratory EFFORT (airway collapse). Central sleep apnea = airflow stops WITHOUT effort (loss of central drive). Cheyne-Stokes respiration is a crescendo-decrescendo periodic breathing pattern of CSA classically tied to congestive heart failure (and also to stroke); other CSA causes include high altitude and opioids.

BOARD TRAP

WRONG answer: prescribing standard CPAP for a heart-failure patient whose study shows Cheyne-Stokes CENTRAL apnea. That is central, not obstructive, disease — management centers on optimizing heart failure (and adaptive servo-ventilation is contraindicated in HFrEF with EF ≤45%). The discriminator: Cheyne-Stokes crescendo-decrescendo pattern with ABSENT effort points to heart failure–related CSA, not obesity-related OSA.
